DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA

LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA

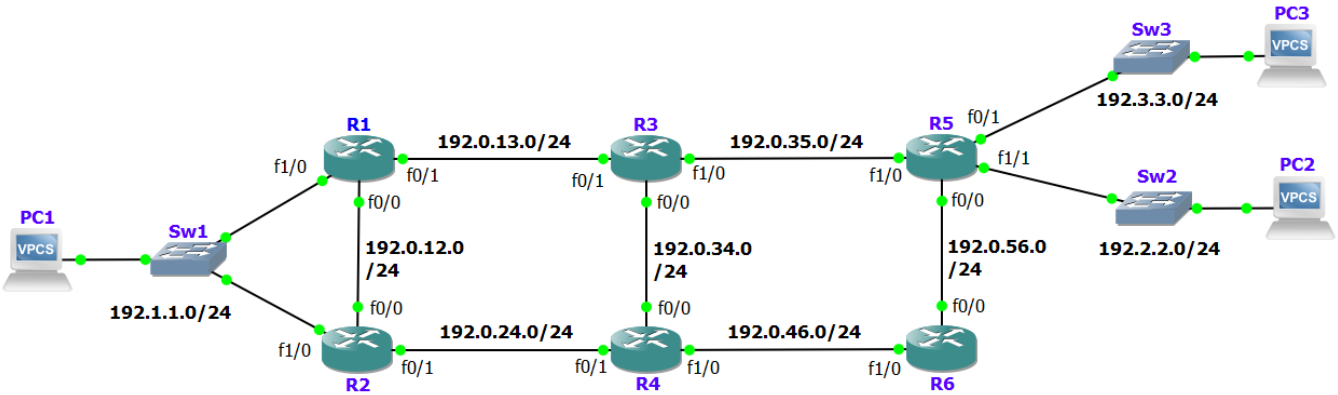
ANO 2025/2026

## REDES DE COMUNICAÇÕES II

# STUDENTS AUTO-EVALUATION OF LABORATORY GUIDE NO. 1

## Scenario 1

Consider the following network with all IP addresses assigned as defined in Laboratory Guide no. 1. Consider that all routers are configured with the RIPv2 protocol, with split-horizon, and all networks except the ones associated with Sw2 and Sw3 are in the RIP processes of the routers. Consider that the IP network of Sw1 is configured in R1 and R2 as a stub network. Consider that VRRP is configured in R1 (set with priority 140) and R2 (set with priority 120) to provide the Default Gateway address 192.1.1.254 to the network of Sw1. Consider that R5 is configured to announce itself to all other RIP routers as the destination of a default RIP route.




Classify as True (T) or False (F) each of the following statements:

- [ ] a) The master virtual default gateway of network 192.1.1.0/24 is R2.
- [ ] b) The IP routing table of R6 has the following RIP entry:

```
R     192.0.13.0/24 [120/2] via 192.0.56.5, f0/0
                    [120/2] via 192.0.46.4, f1/0
```

- [ ] c) The IP routing table of R1 has the following RIP entry:

```
R     192.3.3.0/24 [120/2] via 192.0.13.3, f0/1
```

- [ ] d) The IP routing table of R3 has the following RIP entry:

```
R     192.0.46.0/24 [120/2] via 192.0.34.4, f0/0
```

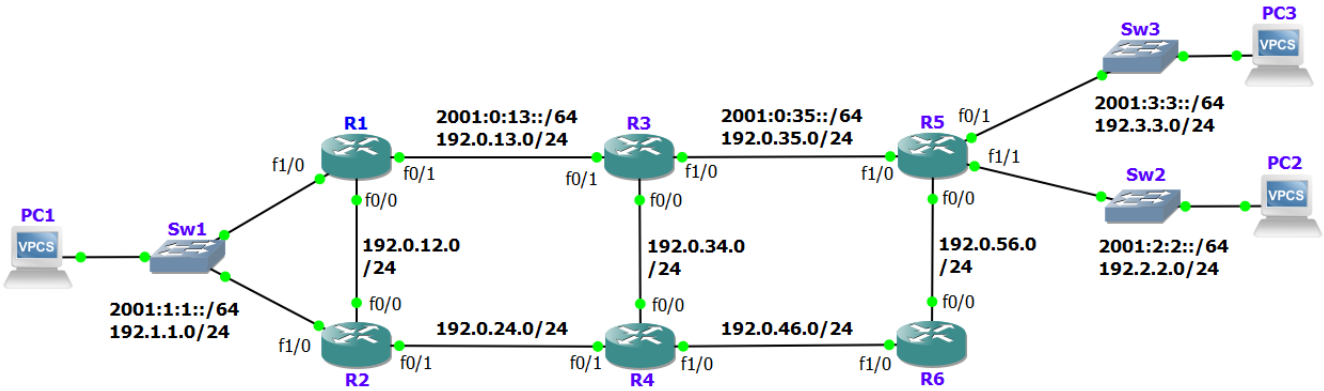
- [ ] e) The IP routing table of R4 has the following RIP entry:

```
R*     0.0.0.0/0 [120/2] via 192.0.34.3, f0/0
                 [120/2] via 192.0.46.6, f1/0
```

- [ ] f) The RIP Response messages sent by R5 through interface `f1/0` include network 192.0.46.0/24 with metric 2.
- [ ] g) The RIP Response messages sent by R1 through interface `f0/0` include network 192.0.46.0/24 with metric 2.
- [ ] h) The RIP Response messages sent by R2 through interface `f0/0` contain 5 IP networks.
- [ ] i) In a ping from PC1 to the address 192.0.24.4, the ECHO Request messages travel through PC1→R1→R2→R4 and the ECHO Reply messages travel through R4→R2→PC1.
- [ ] j) If interface `f1/0` of R3 is shutdown, the RIP entry of R1 to 192.0.46.0 does not change.

## Scenario 2

Consider the following network with all IPv4 and IPv6 addresses assigned as defined in Laboratory Guide no. 1. All routers are configured with one OSPFv2 process in area 0 and routers R1, R3 and R5 are configured with one OSPFv3 process in area 0. In both protocols, all IP networks except the ones associated with Sw2 and Sw3 are included in the OSPF processes and the IP networks of Sw1 are configured as stub networks. R5 is configured to announce itself as the destination of a default route in both protocols. The OSPF costs of all interfaces are 1 in both protocols.




One of the Router Link State entries of the OSPFv2 database is:

```
LS age: 1053
  Options: (No TOS-capability, DC)
  LS Type: Router Links
  Link State ID: 192.0.35.3
  Advertising Router: 192.0.35.3
  LS Seq Number: 80000005
  Checksum: 0x104
  Length: 60
  Number of Links: 3

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 192.0.35.5
     (Link Data) Router Interface address: 192.0.35.3
       TOS 0 Metrics: 1

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 192.0.13.1
     (Link Data) Router Interface address: 192.0.13.3
       TOS 0 Metrics: 1

    Link connected to: a Transit Network
     (Link ID) Designated Router address: 192.0.34.3
     (Link Data) Router Interface address: 192.0.34.3
       TOS 0 Metrics: 1
```

Classify as True (T) or False (F) each of the following statements:

☐ a) The IPv4 routing table of R1 has the following OSPF entry:

```
O    192.0.46.0/24 [110/4] via 192.0.13.3, f0/1
                   [110/4] via 192.0.12.2, f0/0
```

☐ b) The IPv4 routing table of R2 has the following OSPF entry:

```
O    192.0.34.0/24 [110/2] via 192.0.24.4, f0/1
                   [110/3] via 192.0.12.1, f0/0
```

☐ c) The IPv6 routing table of R3 has the following OSPF entry:

```
O    2001:2:2::/64 [110/2] via FE80::C805:16FF:FE40:1C, f1/0
```

☐ d) The IPv6 routing table of R5 has the following OSPF entry:

```
O    2001:1:1::/64 [110/3] via 2001:0:35::3, f1/0
```

☐ e) The database of the OSPFv2 process has 6 Router Link State entries.

☐ f) The database of the OSPFv3 process has 5 Net Link State entries.

☐ g) R5 is the Designated Router of the network 192.0.35.0/24.

☐ h) One of the Net Link State entries has the ID 192.0.13.1.

☐ i) If the IPv4 routing table of R4 has the following OSPF entry:

```
O*E2  0.0.0.0/0 [110/3] via 192.0.46.6, f1/0
                [110/3] via 192.0.34.3, f0/0
```

it means that the default route in R5 was configured as an external route of type E2 with an external cost of 1.

☐ j) If the OSPF cost of interface `f0/1` of R2 is changed to 2 in OSPFv2, the IPv4 routing table of R2 will have the following OSPF entry:

```
O     192.0.56.0/24 [110/4] via 192.0.24.4, f0/1
                    [110/4] via 192.0.12.1, f0/0
```

☐ k) If interface `f0/1` of R3 is shutdown, the IPv4 routing table of R5 will have the following OSPF entry:

```
O     192.1.1.0/24 [110/4] via 192.0.35.3, f1/0
                   [110/4] via 192.0.56.6, f0/0
```